

1. DEPARTMENT: represents academic departments within a college

dept_id | dept_name

1	CSE
2	ECE
3	MBA
4	CSE
5	ME
6	CSE
7	EEE

2. PROGRAM: Defines academic programs and the number of semesters

program_id | program_name | duration_semesters

1	B.Tech	8
2	M.Tech	4
3	MBA	4

3. BATCH: Represents an admission record for a program

batch_id | batch_name | dept_id | program_id

1	2022-26	1	1
2	2022-24	1	2
3	2023-25	3	3
4	2022-26	4	1
5	2021-25	5	1
6	2023-27	6	1

4. SECTION: Master list of section names

section_id | section_name

1	A1
2	A2
3	B1
4	C1

5. BATCH_SECTION: Maps which sections exist in which batch

batch_id | section_id

1	1
1	2
2	1
3	3

4	1
4	2
5	4
6	1
6	2

6. USER: Authentication and authorisation identity

user_id	email	role	password
1	admin_cse@abc.edu	DEPT_ADMIN	#hash
2	admin_mba@abc.edu	DEPT_ADMIN	#hash
3	rao@abc.edu	FACULTY	#hash
4	sharma@abc.edu	FACULTY	#hash
5	admin_cse@xyz.edu	DEPT_ADMIN	#hash
6	kumar@xyz.edu	FACULTY	#hash
7	admin@pqr.edu	DEPT_ADMIN	#hash
8	anita@pqr.edu	FACULTY	#hash

7. FACULTY: Academic identity of a faculty member

faculty_id	user_id	dept_id
10	3	1
11	4	1
12	6	4
13	8	6

8. FACULTY_BATCH_SECTION: Defines which sections faculty deal with

faculty_id	batch_id	section_id	subject_code
10	1	1	cs302
10	1	2	cs301
11	2	1	cs308
12	4	1	cs401
12	4	2	cs506
13	6	1	cs703

9. STUDENT: Stores the student's academic identity

student_id	roll_no	batch_id	section_id	dept_id	email	phone
1	22CS101	1	1	1	#email	#phone
2	22CS102	1	1	1	#email	#phone
3	22CS201	1	2	1	#email	#phone
4	22MT101	2	1	1	#email	#phone
5	23MBA01	3	3	3	#email	#phone

6	22CSX01	4	1	4	#email	#phone
7	22CSX02	4	2	4	#email	#phone
8	21MEX01	5	4	5	#email	#phone
9	23CSP01	6	1	6	#email	#phone
10	23CSP02	6	2	6	#email	#phone

10. SEMESTER: Tracks semester progression per batch

semester_id	batch_id	semester_no	is_active

1	1	5	true(present semester for that batch)
2	2	3	true
3	3	1	true
4	4	6	true
5	5	8	true
6	6	1	true

11. ATTENDANCE_SESSION: Represents a single class conducted

session_id	semester_id	batch_id	section_id	faculty_id	date	subject_code

1001	1	1	1	10	2025-02-01	CS301
1002	4	4	1	12	2025-02-01	CS306
1003	6	6	1	13	2025-02-01	CS305

12. ATTENDANCE_RECORD: Stores student presence per session

session_id	student_id	status

1001	1	P
1001	2	A
1002	6	P
1002	7	P
1003	9	A